

VATER RULES — the numbered rulebook (compiled 2026-08-12)

Every standing rule Trey/Jared has given for how Vater videos are created and operated, in one place, numbered for quick reference ("rule 47").

Numbering doctrine (same as the library #N system): rule numbers are PERMANENT — never renumbered, never reused. New rules get the next free number and are appended to their section. Retired rules keep their number, struck through with a date.

Authority chain: machine truth is `VATER-STANDING-SPEC.json` (read by `vater.py` and the finance-pipeline harnesses via `finance-pipeline/standing-spec.mjs`). This file is the human-readable layer. **To change a rule:** edit or annotate it here — the next Vater session syncs the change into `VATER-STANDING-SPEC.{md,json}` (and `VATER-SETTINGS.env` where applicable). Each rule cites its source so drill-down is one hop.

1. Intake, script & approval gates

1. **Script-first.** The source of truth is TREY'S SCRIPT. Reference videos are RETIRED — no yt-dlp, no whisper transcription, no LLM scripting pass, and no spend of any kind until Approve. (*VATER-STANDING-SPEC.md §2*)
2. **Script Review flow:** Trey pastes/uploads → reads/edits/saves → **Approve & Animate**. New rows park at `awaiting_script_approval` via `POST /api/vater/youtube/from-script`. (*vater-script-first-jeff-whitfield*)
3. **Nothing renders before approval.** `stopAfterScript` → `script_ready` → `awaiting_script_approval` → re-kick with `scriptOverride`. (*vater-video-qa-doctrine*)
4. **Script saves are versioned, append-only** (`YouTubeProject.scriptVersions`: generated/edited/approved, cap 20). (*vater-standing-spec*)
5. **Draft-gated output.** No publishing, no YouTube upload, no customer handoff without Trey/Jared pulling the trigger. (*vater-sandbox-doctrine, vater-direct-lane*)
6. **Fact-check the script stage.** Kimi hallucinated "avg card debt \$1,000" (real ≈ \$6.5k) — numbers in scripts get verified. (*vater-finance-pipeline*)
7. **Runtime comes from the script, never from stretching scenes.** Want a longer video → write more words. Target words ≈ minutes × 185 (Monroe long-form). (*vater-video-qa-doctrine, SPEC §6*)
8. **A missing locked style is a hard 409, never a silent fallback.** (*vater-script-first-jeff-whitfield*)
9. **DO-NOT-SHIP: "wealth-in-30s"** — video + short stay in the library; no publish/upload/staging until Jared reverses. (*vater-finance-pipeline*)
10. **#21 "Buy vs Rent"** (`cmsoxr9od0001l4ng7geu9hqo`) — rejected by Trey 8/12; repair SHIPPED 8/13 (93 slides remade against benchmark #19, recomposed to the same blob key, \$13.27 all-in). Captions/wpm fixes still open; publishing still waits on Trey/Jared per rule 5. (*vater-library-numbering*)

11. **Course lane gates:** Approve & Render in Course Studio is the structural gate; hard STOP after the Lesson-1 pilot for quality+cost review before any batch. (*vater-financial-course*)

2. Style & look

12. **ONE locked style: "Jeff Whitfield 3-D style"** (formerly VATER-3D / Animate2-Modal). The site's style picker is GONE; Script Review binds to this row via `lib/vater/locked-style.ts` (DB `cmofyvao30000l204op279f52`). (*SPEC §1*)
13. **Stills** = preset `pixar` + custom art style "**Finance Pixar 3D**" (`finance-pixar-test`). (*SPEC §1*)
14. **NEVER 2D.** Persistent default, not a per-job argument. (*SPEC §1*)
15. **NEVER photoreal / live-action / realistic humans** — every scene stays stylized 3-D cartoon. (*SPEC json neverPhotoreal*, Trey 8/10)
16. **Motion engine:** Wan2.2 **Animate-2 on Modal** (`vater-wan22.animate2_clip`, L40S, distill Q8, 6-step, 16fps native → 32fps compose, 81 frames/chunk max); fallback `modal-wan22` for scenes with no usable driver. (*SPEC §1*)
17. **Every still scene gets camera motion — never a frozen frame.** Four directions (zoom in / zoom out / pan L → R / pan R → L) cycling so consecutive scenes never repeat; smooth, no jitter; 4× supersample before zoompan and don't lower it (`ZOOM_MAX 1.12`, `PAN_ZOOM 1.08`). (*SPEC json stillMotion*, Trey-approved 8/10)
18. **Default cost posture = STILLS + PANNING; animation is opt-in per video, never the default** (still ≈ \$0.05 vs animated clip ≈ \$0.50). (*SPEC json costPosture*, Trey 8/10)
19. **Hard cuts only — no fading between scenes** (`VATER_SCENE_XFADE_S=0`, tail trim 0.20s); keep the single 1.5s end fade. (*vater-video-qa-doctrine*, *VATER-SETTINGS.env*)
20. **Quality is locked** at 848x480 @ 32fps, NVENC p4 cq23 (~1.1 Mb/s), libx264 crf20 fallback, AAC 192k 48kHz, 4× still supersample, no frame interpolation. Changes need explicit approval. (*SPEC §8*)
21. **Captions are LOCKED ON exactly as they are** — style, size, position, timing untouched without Trey's explicit approval. (*SPEC json captions*, Trey 8/10)
22. **Animated clips need a camera push** (`VATER_ANIM_PUSH=0.12`) or they read as slides; judge "does it look animated" by within-scene frame diff 2s apart, never across a cut. (*vater-video-qa-doctrine*)

3. Cast & characters

23. **Host = Jeff Whitfield, in every video**, the primary character explaining the finance content. Assets `characters/jeff-whitfield/`; seed 7 (`VATER_BASE_SEED`); no LoRA — identity = descriptor + pinned seed + `cast_match QA`. (*SPEC §3*)
24. **Closed cast.** The only humans allowed anywhere: Jeff Whitfield, Linda Whitfield, David Reyes, Cooper Shaw, Julie Tran, Lynette Brooks — and only in scenes the script names. No tellers, colleagues, passers-by, crowds, background figures, or unnamed hands. A scene naming no roster character has NO people at all. (*SPEC json castRules.closedCast*)
25. **Script-named roster characters MUST appear; unnamed ones must not.** `lockRoster` turns auto-casting off. (*SPEC json characterLock*)

26. **NEVER two of the same character in one frame** (no Jeff looking at Jeff) — enforced in planner prompt, deterministic lint, and the `no_clones` QA check. Two *different* characters sharing a frame is fine. (*SPEC §3c*)
27. **ONLY A HUMAN HAS A FACE.** Piggy banks, cards, calculators, jars, coins are inanimate — never eyes/mouths/limbs, never cute or anthropomorphized. (*SPEC json facesRule*)
28. **No faced objects for counts** — never piggy banks / moneyboxes / mascots / figurines (proven unfixable); use bank cards, passbooks, labelled glass jars, envelopes, statements, cash stacks, on-screen account cards. Enforced by `_FACED_OBJECT_RE`. (*SPEC json noFacedObjects*)
29. **Identity lock.** Every character keeps identical face, hair (color/length/style), facial hair, eye color, skin tone, freckles, earrings, and build in every scene of every video. **ONLY** clothing and setting vary. (*SPEC json characterLock, Trey 8/10*)
30. **Per-character invariants are explicit:** David = tightly-curved black hair, always bearded; Cooper = copper-ginger wavy hair, always bearded, green eyes, freckles; Julie = always high ponytail, silver studs, no freckles; Lynette = shoulder-length cornrow braids, gold hoops. (*SPEC json supportingCharacters*)
31. **Descriptors state locked features as POSITIVE concrete description, never negations** — "never down" produced hair down. (*SPEC json descriptorWritingRule*)
32. **Wardrobe = one outfit per ACT** (up to 3 acts/video), seeded by (character, jobId) so resumes re-pick the same outfit; never per-scene randomness. Clothing was removed from identity descriptors so it can vary. (*SPEC §3a*)
33. **Jeff is a financial businessman — suits or professional attire, ALWAYS.** Palette characters/jeff-whitfield/wardrobe.json. (*SPEC §3a*)
34. **The host stays index 0 in any roster** (vater.py treats characters[0] as host); Prisma character includes have no ORDER BY — sort oldest-first. (*vater-script-first-jeff-whitfield*)
35. **NEVER clone a real person's voice or use real-person photo refs.** (*vater-video-qa-doctrine*)

4. Scenes, backgrounds & on-screen text

36. **Backgrounds MUST VARY:** ≥ 3 distinct settings per video, no single setting over $\sim \frac{1}{3}$ of scenes, interiors + exteriors mixed; the narration picks the place. Palette SETTINGS-PALETTE.json. (*SPEC §4, Trey-approved 8/10*)
37. **On-screen text MINIMAL — numbers only**, matching what the script is saying at that moment; never paragraphs, headlines, signs, labels, or characters holding written text. (*SPEC §7*)
38. **\$-header / drawtext cards stay OFF** (VATER_FINANCE_HEADERS=off, Trey 8/8). Manual editor overlays still render. (*vater-video-qa-doctrine*)
39. **KEEP and encourage charts, graphs, ledgers, calculators, and financial tools Jeff is using** — clean and near-unlabeled. More financial imagery overall. (*SPEC §7*)
40. **More financial props per scene, but props carry little to NO readable text** (22-item palette in SPEC json propsPalette). (*Trey 8/10*)
41. **Props must be motivated by that beat's narration. NO symbolic/surreal props ever** (brain models, word-globes, badges, trophies) — `_strip_surreal_props`. (*vater-video-qa-doctrine*)
42. **Visuals land in the SAME scene as their narration — never one beat late.** (*vater-video-qa-doctrine*)

43. **Expressions match the narration's emotional tone per scene** (planner emits `emotion` from a closed vocabulary). (*vater-video-qa-doctrine*)
44. **Visual vocabulary rule:** when the narration states a COUNT, NUMBER, LIST, COMPARISON, or FLOW — SHOW it, usually without the host ("eight bank accounts" → eight countable cards; "4.9% vs 0.01%" → both figures large, side by side). (*SPEC json visualVocabulary*)
45. **At least 1 scene in 3 is a people-free financial visual** — and these visuals are the ONE place on-screen text (figures, short labels, list lines) is encouraged. (*SPEC json minimumShare/textException*)
46. **BIG props:** a screen or phone filling about half the frame, shot close, clearly readable, showing the figure the narration just said. Digits render reliably; long words render as gibberish. (*SPEC json bigProps/textOnProps*)
47. **One labelled container (jar) per named fund** the narration mentions. (*SPEC json labelledContainers*)
48. **Banned filler poses:** pointing off into the distance, two fingers / peace sign / thumbs up at camera, staring at a wall calendar, gazing at an unspecified paper. (*SPEC json bannedFillerPoses*)
49. **Host in ~half of all scenes**, doing something that matches the narration; the other half people-free financial visuals. (*SPEC json hostPresence*)
50. **Never a scene that contradicts the narration** (broken car window with no car in the script; Christmas ornaments during sinking funds). (*SPEC json benchmarkVideo.fix*)
51. **Strip anatomy clauses from the style suffix on people-free scenes** (`_scene_style_suffix`) — that suffix is what put faces on objects. (*SPEC json rootCause2026_08_10*)

5. Voice & audio

52. **Voice = Monroe** — local IndexTTS-2 clone (`vater_voices/Monroe.wav`) served via Modal app `vater-tts` (`voiceBackend: "indextts-modal"`). (*SPEC §6, vater-modal-only*)
53. **The TTS Modal path NEVER falls back to f5-tts** — that fallback silently re-enters the local GPU. (*vater-modal-only*)
54. **Approved tempo:** `VATER_TTS_INTERVAL_MS=340`, `VATER_TTS_CHUNK_GAP_MS=320`, `VATER_TTS_TEMPO=0.94` (~155 wpm delivered). (*SPEC json voice.pacing*)
55. **Measure wpm per voice** — Monroe reads ≈185 wpm long-form / ≈164 short; short samples mislead. (*SPEC §6, vater-video-qa-doctrine*)
56. **Tone = calm, authoritative, human, unhurried — explaining, not selling** (Nick Invests feel). (*vater-video-qa-doctrine*)
57. **Punctuation IS the tempo.** Scripts are written punctuation-for-the-ear; an em dash produces a long beat in IndexTTS-2. The ref clip's own pace is the real lever; sampler/emotion params barely move it (±3 wpm). (*vater-video-qa-doctrine, vater-finance-pipeline*)
58. **Dollar amounts spelled out in words in the script AND voiceover** ("eight hundred thirty-nine thousand dollars", never "\$839k"); `speakable.py` handles legacy digit scripts. ⚠ See conflict C7 (captions want numerals). (*vater-video-qa-doctrine*)
59. **Acronyms spelled out in the spoken read only:** IRA → "I-R-A", FDIC → "F-D-I-C" (`_spell_finance_acronyms`); captions keep "Roth IRA". (*SPEC json pronunciation*)

60. **Lip sync OFF.** Mouths neutral or closed; S2V stays per-scene ask-only; don't spend compute on it. (*SPEC §5*)
61. **Final mux pins -ar 48000 -ac 2.** (*vater-blob-delivery*)

6. QA & pre-ship

62. **Still-QA runs on every still AFTER stills, BEFORE Modal animation spend** (*vater_scene_qa.py*, Gemini vision): no_clones, person_count_ok, cast_match, props_motivated, no_readable_text, anatomy_ok, no_ghost_figures, setting_props_consistent, prop_type_match, no_self_regard. **VATER_SCENE_QA=on.** (*vater-video-qa-doctrine*)
63. **Clip-QA** (*vater_clip_qa.py*) judges 3 frames per animated clip (30/60/90%) before compose; fail → one re-animate → revert to Ken Burns still. (*vater-video-qa-doctrine*)
64. **Final pre-ship step is a human/Claude EYEBALL of extracted frames (60% + 90%) of every animated clip** — Gemini flash misses subtle extra hands at 480p and false-flags cartoon finger counts. (*vater-video-qa-doctrine*)
65. **Always extract frames from the FINAL at the fixed scenes' timestamps to confirm a revision landed** — never trust "REVISED + RECOMPOSED". (*vater-video-qa-doctrine*)
66. **Gate QA on defects you can point at;** expression_ok is ADVISORY only. Re-roll budget: 2 per scene, 20 global. (*vater-video-qa-doctrine*, *VATER-SETTINGS.env*)
67. **Always eyeball crowd / "most people" beats** — they render as 2–5 identical clones and still-QA passes them; fix via single-host or prop staging. (*vater-standing-spec*)
68. **Never stage prop-only beats without the host** — FireRed grows faces on prop-only scenes. (*vater-standing-spec*)
69. **Production-ready standing order:** every video is built to be publishable without Jared re-reviewing — self-check anatomy, numbers, timing, tempo before delivering. (*vater-video-qa-doctrine*)
70. **Trey's/Jared's scene numbers in reviews are 1-BASED** (his "scene 3" = scenes/002.png). (*vater-video-qa-doctrine*)
71. **Motion check:** average motion score < 15 (mean pixel diff) → STOP, the prompts didn't unlock motion. Target 15–25 for narrative. (*vater-animation-recipe*)
72. **Never kick parallel /animate-scene calls for the same scene index** (version-numbering race, unfixed). (*vater-animation-recipe*)

7. Backend & infra

73. **HARD RULE: Vater is MODAL-ONLY. Never run any part of a Vater render on the DGX GPU.** ("never use Vater with GPU only send it to Modal" — Jared 8/10) (*vater-modal-only*)
74. **NEVER systemctl --user start/restart comfyui-vater** (or any ComfyUI) to unblock a Vater job — route it to Modal instead. (*vater-modal-only*)
75. **All four lanes are Modal:** stills firered-modal, animation modal-wan22, planner/script LLM off-box via LiteLLM :4000 → fallback/kimi-k2-turbo, voice indextts-modal. (*vater-modal-only*)
76. **wan22-local on the GB10 is explicitly rejected** (15–20 min/clip — "a 30 sec clip that takes 3 hours is not practical"). (*vater-finance-pipeline*, Jared 8/5)

- 77. **Runway-clear / stopping other prod services gates on real local-GPU use (uses_local_gpu)**, never on videoBackend == "sdxl"; keep the flag and schedule_restore in sync. (vater-modal-only)
- 78. **One general env-driven harness: ~/vater-studio/finance-pipeline/vater-render.mjs** — no more fork-per-video scripts. (vater-modal-only)
- 79. **systemctl --user restart content-autopilot-api** RESUMES every job stored as **running** — before restarting, check vater_jobs.json and POST /vater/cancel-job on each; gate the restart on the check output, never chain check && restart in one command. (vater-script-first, vater-financial-course)
- 80. **Never default to sdxl-local / sdxl-ipadapter / FLUX for scene gen.** (feedback_vater_no_sdxl)
- 81. **Modal wan batches chunk at VATER_ANIM_CHUNK_SCENES=12; a DGX cancel does NOT kill Modal tasks** — the function timeout is the only real kill. (vater-finance-pipeline)
- 82. **The poll route's done handler must MERGE scenesJson per idx, never overwrite** (overwrite wipes mediaType/videoUrl/videoVersion/animQuality). (feedback_vater_poll_merge_scenes)
- 83. **Vater Watcher runs on a scoped allowlist** (5 MCP tools + Read/Grep/Glob, no Bash/Edit/Write); real changes only via propose_action → user clicks Apply. (vater-watcher)
- 84. **Long-form renders use a 15-min AbortSignal on fetch** (cold Modal regen exceeds undici's 5-min default). (vater-standing-spec)
- 85. **Never hardcode style/character/settings in a harness** — import finance-pipeline/standing-spec.mjs or read the SPEC.json; any spec change lands in VATER-STANDING-SPEC.{md,json} FIRST. (vater-standing-spec)
- 86. **Never hardcode secrets** — set -a; . ~/vater-studio/VATER-SETTINGS.env.
CONTENT_API_KEY truth = the content-autopilot-api systemd unit; never a code default; never in a client bundle. (content-api-key-rotation)

8. Library & delivery

- 87. **Permanent #N — title prefixes** on every library video (per owner, oldest-first); numbers NEVER change or get reused — Trey references videos by number. Legacy ownerless rows use #LN — . Enforced by lib/vater/video-number.ts + lazy sweep on every library fetch. (SPEC §9)
- 88. **Never create or rename a Vater video without keeping the #N — prefix intact.** (vater-library-numbering)
- 89. **The approved format benchmark is #19 — The Three-Account System (v3)**, project cmsolefqb0001l446nhl1psbl (Trey corrected the pointer 8/12; his 8/11 approval was for v3). Never cite the benchmark by title alone — four renders share it (#15 v1, #18 v2, #19 v3 = APPROVED, #20 v4). (SPEC §9 + json benchmarkVideo)
- 90. **Every render/example goes to the studio Library + green NEW banner push (/api/vater/latest)** — never NAS-only. (vater-sandbox-doctrine)
- 91. **Finals serve from Vercel Blob CDN** (vater-finals/<projectId>.mp4, ?v=<ts>); playback links straight to *.blob.vercel-storage.com via finalVideoPlaybackUrl(), never the serverless proxy. Download stays on the proxy (?download=1). (vater-blob-delivery)

- 92. **The poll route must NEVER downgrade an https finalVideoUrl back to a proxy path;** any new run script writes the blob URL. (*vater-blob-delivery*)
- 93. **Compose always writes -movflags +faststart** (moov at front). (*vater-blob-delivery*)
- 94. **Audio-but-no-video / stutter for external users is a DELIVERY problem** — never re-litigate codecs for that symptom class. (*vater-blob-delivery*)

9. Cost, billing & payments

- 95. **The per-video all-in total is part of the deliverable** — never report a video done without its cost; never make Jared ask. (*vater-cost-truth-automated*)
- 96. **Costs are auto-reconciled from the job record** (`scripts/vater-cost-reconcile.mjs`, cron every 20 min, log `~/vater-studio/cost-reconcile.log`) — never hand-total a video. (*vater-cost-truth-automated*)
- 97. **DB byJob is the durable ledger; file amounts MERGE over it, never rebuild.** A total going DOWN with fewer jobs = rotated history, not a correction. (*vater-cost-truth-automated*)
- 98. **Never hand-edit costJson without adding the jobIds to byJob.** (*vater-cost-truth-automated*)
- 99. **Apply VATER_MODAL_OVERHEAD_MULT=1.6 at cost finalize;** if reconciliation still gaps, tune the mult — don't stack rows. (*VATER-SETTINGS.env*, *vater-finance-pipeline*)
- 100. **Invoicing = exactly two lines:** Compute (at cost, passed through unchanged) + Render Operations (finished minutes × rate). (*vater-billing-ops-fee*)
- 101. **VATER_OPS_RATE_PER_MIN = 0.35;** every code default must equal 0.35 — NEVER reintroduce a 1.00 fallback. (*vater-billing-ops-fee*)
- 102. **ONE NUMBER on the tab** = Compute (all cash, at cost) + Render Operations; click-through shows the FULL provider breakdown as plain one-liners — never strip the provider list, no parentheses, no rate math. (*vater-billing-ops-fee*)
- 103. **A running total must NEVER shrink** — the pill shows \$X spent · \$Y billed. (*vater-billing-ops-fee*)
- 104. **Token counts are DEAD on all cost surfaces** — dollars and stage breakdown only. (*vater-billing-ops-fee*)
- 105. **Cost dollars = REAL cash only** (Modal/Gemini/fal); Claude tokens never convert to dollars. (*vater-finance-pipeline*)
- 106. **Local-first cost discipline:** DGX \$0 → Modal cents → FLAG before any new paid API (incl. fal.ai). (*vater-sandbox-doctrine*)
- 107. **Billing/payment totals math lives ONLY in lib/vater/billing/summary.ts** (`getVaterBillingSummary`); never fork it. (*vater-payments-due-reset*)
- 108. **recordVaterPayment()** is the ONE write path for payments (a bare `prisma.vaterPayment.create` loses the snapshot). (*vater-payments-due-reset*)
- 109. **All-time total NEVER resets; dueUsd = totalUsd – payments.** Reset when the weekly Zelle lands, via `/hq "Trey render bill"` or `node ~/vater-studio/vater-payment.mjs <amt> --note "..."`. (*vater-payments-due-reset*)
- 110. **Never record a payment before the Zelle actually lands** (hard-money rule). (*vater-payments-due-reset*)

111. **The ops fee is REPORTING only** — live Stripe charging is untouched; wiring it needs Jared's explicit word. *(vater-billing-ops-fee)*
112. **The cost pill stays all-in across BOTH lanes** (YouTube + course) — never filter the `/api/vater/latest` rollups. *(vater-financial-course)*

10. Scope & sandbox

113. **"Vater" scopes a conversation EXCLUSIVELY to the /animate studio + its plumbing** — nothing else gets changed, added, edited, or even verified. *(vater-sandbox-doctrine)*
114. **In-scope surface:** `tolley.io/animate`, `app/api/vater/*`, `components/animate/**`, `components/vater/**`, `lib/vater/**`, DGX content-autopilot-api :8096 vater routes, `comfyui-vater:8189` (dormant per rule 73), `vater-priority`, `claude-vater-watcher`, Modal apps `vater-wan22/vater-firered/vater-tts`, `~/vater-studio/`. *(vater-sandbox-doctrine)*
115. **Broken non-Vater things get FLAGGED to Jared, never fixed inside a Vater conversation.** *(vater-sandbox-doctrine)*
116. **Trey's YouTube OAuth tokens go to vater-scoped storage** (`SocialAccount` via `/api/vater/social-accounts/oauth/youtube/*`) — NEVER `PlatformConnection` (Jared's channels). *(vater-sandbox-doctrine)*
117. **Studio-only access = VATER_STUDIO_ALLOWLIST_EMAILS only** — NEVER `VATER_ADMIN_ALLOWLIST_EMAILS` (that flag exposes Plaid/net-worth data; live breach 2026-08-05). *(vater-studio-access-tier)*
118. **Verify access changes against prod env** (`npm run vercel env pull`), never local `.env`. *(vater-studio-access-tier)*
119. **The Direct lane runs as Unix user vater** (uid 1001, no sudo/docker, ACL-sealed to `~/vater-studio`); all renders via the :8096 API; finance scripts stay owner-side. *(vater-direct-lane)*

Appendix: Known conflicts & stale pointers (need a ruling or cleanup)

- **C1 — Benchmark #19 v3 vs #20 v4.** SPEC md+json (newest write, 8/12 18:39) say **#19 v3** (`cmsolefq0001l446nhllpsbl`) and document Trey's 8/12 self-correction (his 8/11 approval was v3). Two memory files and the JSON's top-level `authority` string still said #20 v4 — memory now corrected; the JSON `authority` line is stale and awaits a spec-sync pass.
- **C2 — Standing spec v1 vs v2.** v1 (host "Ray Whitfield", reference videos, style picker) is fully superseded by v2 (Jeff Whitfield, script-first, picker gone, wardrobe/backgrounds vary).
- **C3 — Reference-video residue.** SPEC json still carries `referencePolicy: "subject-only"`; the reused-reference 409 gate and `script-from-reference` route exist in code but are unreachable from the UI. Vestigial under v2.
- **C4 — VATER-SETTINGS.env still has FINANCE_CHARACTER_NAME=Ray Whitfield.** Possibly load-bearing in prompts — needs a verify pass + Trey's ok before changing (output-affecting). Driver dir already repointed to jeff-whitfield.

- **C5 — Linda's status.** SPEC md §3b says "pending critique"; json + REVISION-LOG say LOCKED (Trey 8/10, "Linda looks good"). The md paragraph is stale; David/Cooper/Julie/Lynette exist only in the json, not the md.
- **C6 — On-screen numbers have no mechanism.** Cards are OFF (rule 38) but the approved format wants the narration's numbers on big props — today that depends entirely on the image model. Re-enabling Remotion `counter_roll/stat_card_pop` needs Trey's yes.
- **C7 — Captions numerals vs words.** REVISION-LOG Round 1 locked captions to numerals (\$6,500) with `speakable()` for the TTS read only; QA doctrine rule 5 wants dollar amounts in WORDS in the script. Directly opposed — documented root cause of #21's mangled captions ("ninety ,000"). Needs one ruling.
- **C8 — Caption style karaoke vs phrase blocks.** April doctrine said phrase blocks, "NEVER karaoke"; current pipeline ships karaoke captions, now LOCKED ON by Trey (rule 21). Current lock wins; also open whether "not a lot of text" covers captions — ASK BEFORE CHANGING.
- **C9 — Background variety vs story continuity.** The planner's STORY-CONTINUITY rule beats the ≥ 3 -settings rule on continuous scripts; continuity currently wins. Variety rule likely needs scoping per-act.
- **C10 — Kling/wan hybrid dormant.** The 8/8 Kling-first animation doctrine is dormant (not deleted) under the 8/10 stills-default cost posture; `kling-pro` is dead in code.
- **C11 — Transitions.** Round 2's 0.3s crossfade was reversed 8/8 → hard cuts only (rule 19). Hard cuts current.
- **C12 — Scene-gen backend.** April's `firered-local` default is superseded by `firered-modal` under Modal-only (rule 73).
- **C13 — Voice lineage.** Sterling → Wallace → **Monroe** (8/8). Any doc naming Sterling/Wallace as the finance voice is superseded.
- **C14 — wpm figures disagree** (185 long-form vs ~155 delivered at TEMPO 0.94 vs 164 short) — runtime estimation is the live weak spot; #21 was planned at a stale 155 while Monroe ran ~180.
- **C15 — vater.py still prefixes prompts with "A 2D cartoon male character."** Deliberately untouched — it produced the look Trey approved; changing it counts as a quality change needing approval. The prefix drops when a named secondary shares the frame.